

06bis. Prototyping

Interazione Uomo Macchina
Prof. Maurizio Mancini



DIPARTIMENTO
DI INFORMATICA
SAPIENZA
UNIVERSITÀ DI ROMA

Human Computer Interaction
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<https://github.com/polito-hci-2022>



The Goal

- **Envisionment:** making ideas visible
 - Generating new ideas
 - Evaluating new ideas (within the design group)
 - Testing new ideas (with users)
- Different tools and techniques, according to
 - The stage of design (early, ..., advanced, final)
 - The intended audience (designers, test users, clients, management, ...)
- **Error to avoid:** focusing on the user interface before focusing on the task that the user has to accomplish

Techniques

- Sketches
- Maps
- Low Fidelity (paper) prototypes
- Video Prototypes
- Medium Fidelity Prototypes
- High Fidelity Prototypes

“If a picture is worth a thousand words, a prototype is worth a thousand meetings” — IDEO

Low Fidelity Prototypes

How to start using an application, months before implementing it

Paper Prototypes

- A hand-drawn mock-up of the user interface (usually) on multiple sheets of paper of varying sizes



Key Features for Paper Prototypes

- Interactive paper mockup
 - Sketches of screen appearance
 - Paper pieces show windows, menus, dialog boxes
- Interaction is natural
 - Pointing with a finger = mouse click
 - Writing = typing
- A person simulates the computer's operation
 - Putting down & picking up pieces
 - Writing responses on the “screen”
 - Describing effects that are hard to show on paper
- Low fidelity in look & feel
- High fidelity in depth (person simulates the backend)

Materials

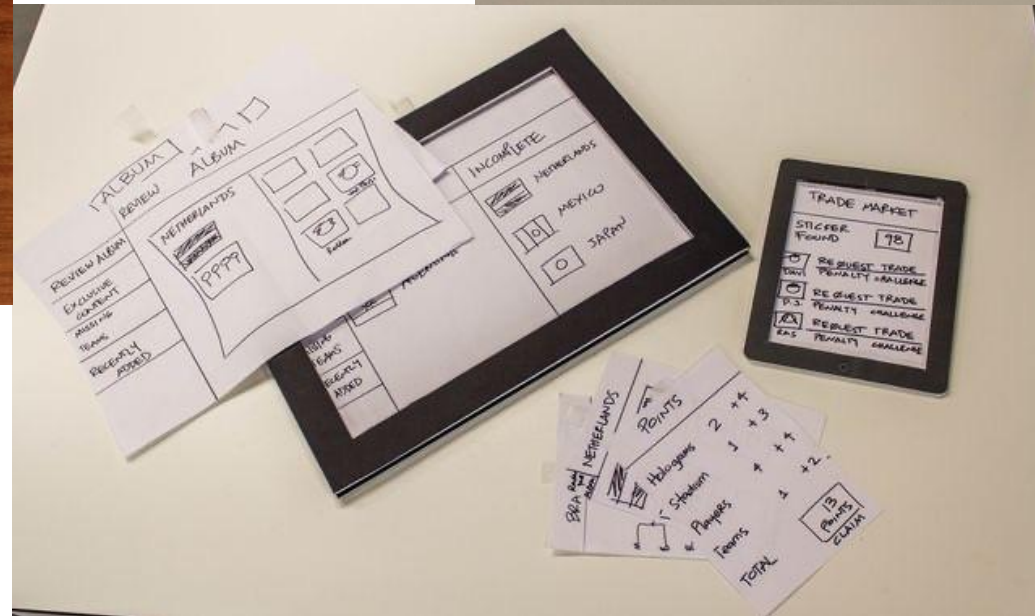
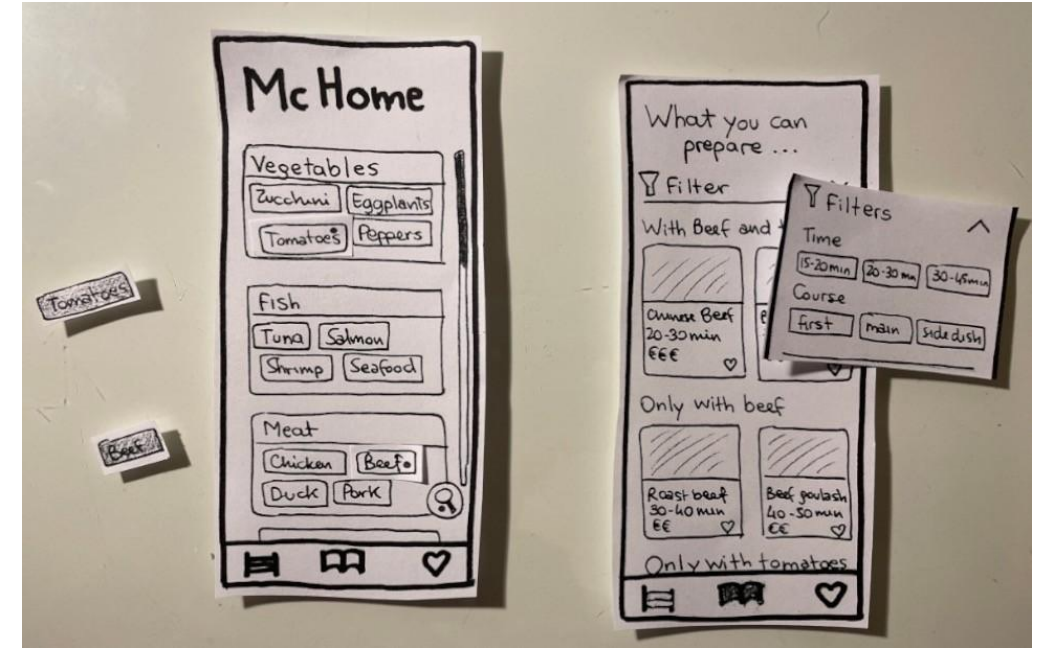
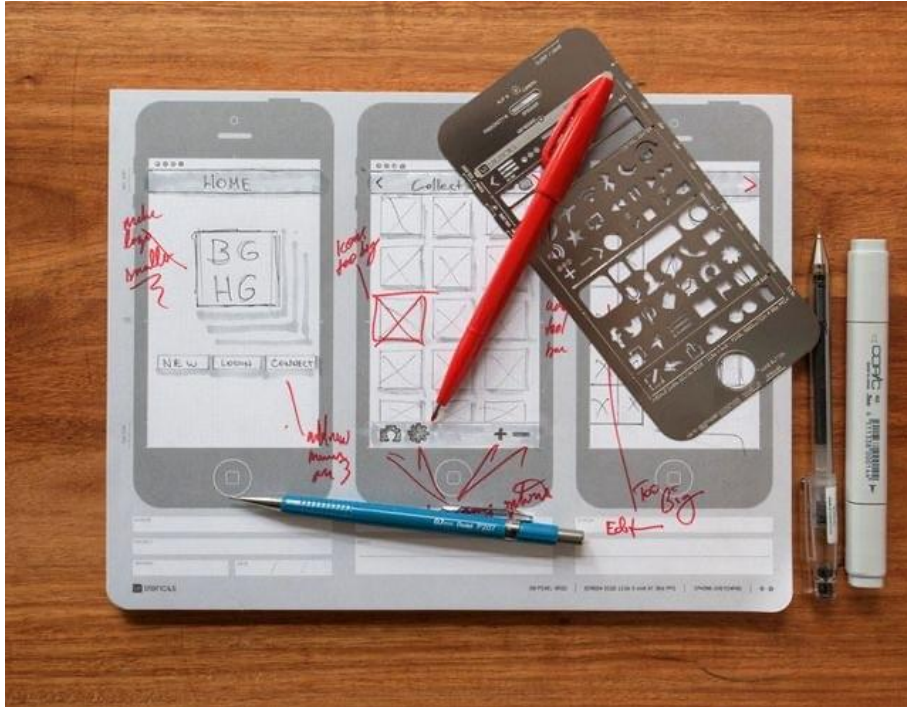
- Paper, Transparent paper
- Pens, Markers
- Post-It notes
- Glues, scotch tape, scissors
- Photocopies
- UI Stencils
- Reusable UI components
- Printouts of screenshots



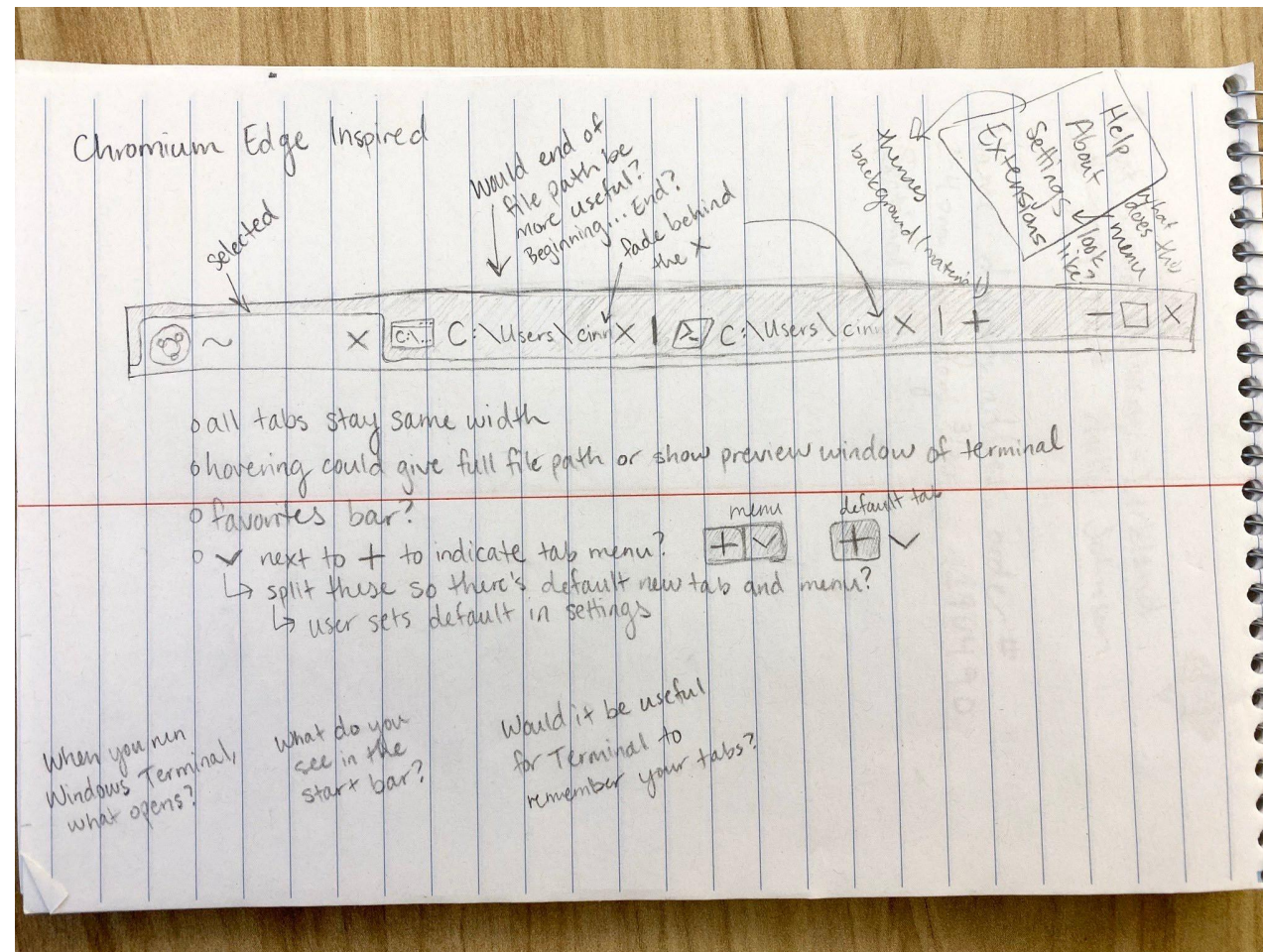
Why Paper Prototyping?

- Faster to build
 - Sketching is faster than programming
- Easier to change
 - Easy to make changes between user tests, or even **during** a user test
 - No code investment - everything will be thrown away (except the design)
- Focuses attention on big picture
 - Designer doesn't waste time on details
 - Customer makes more creative suggestions, not nitpicking
- Non-programmers can help
 - Only kindergarten skills are required

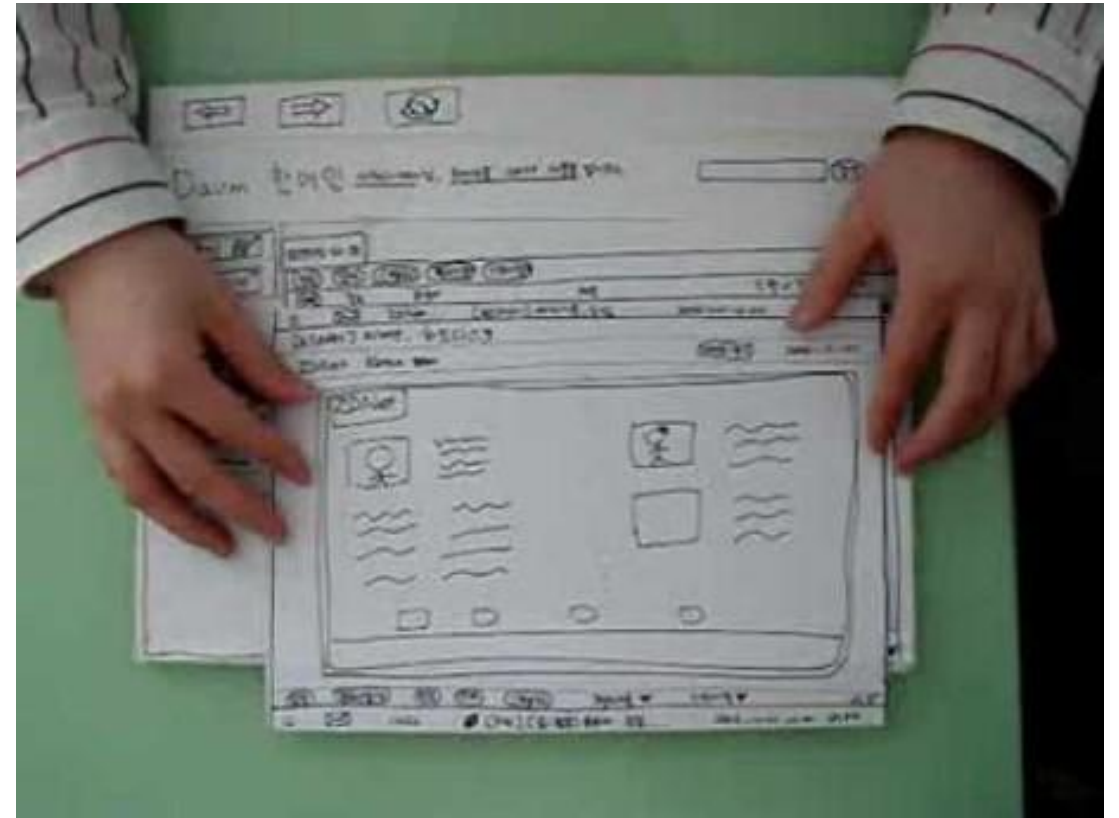
Paper Prototypes: Examples



First Ever Mockup of the Windows Terminal Tab Bar

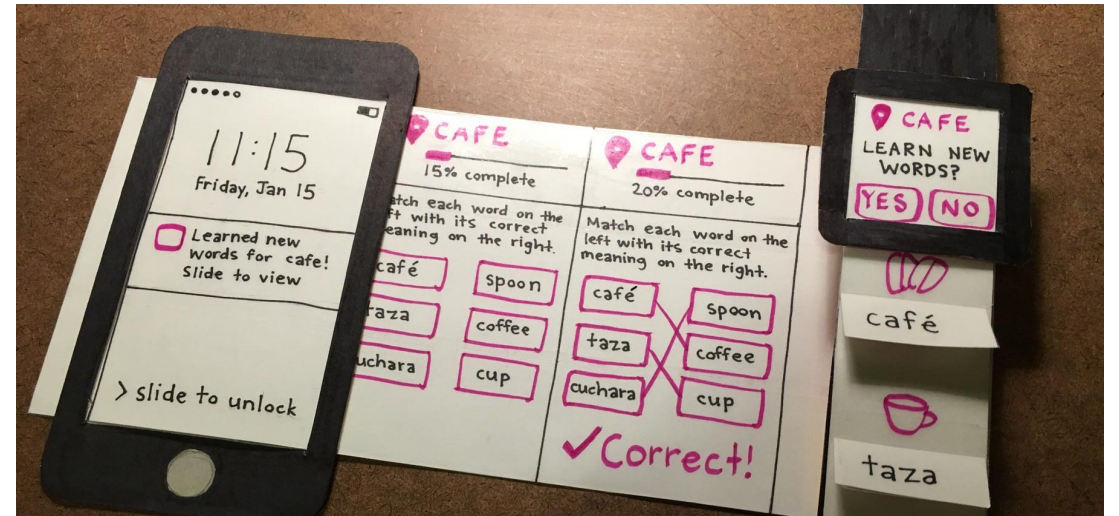
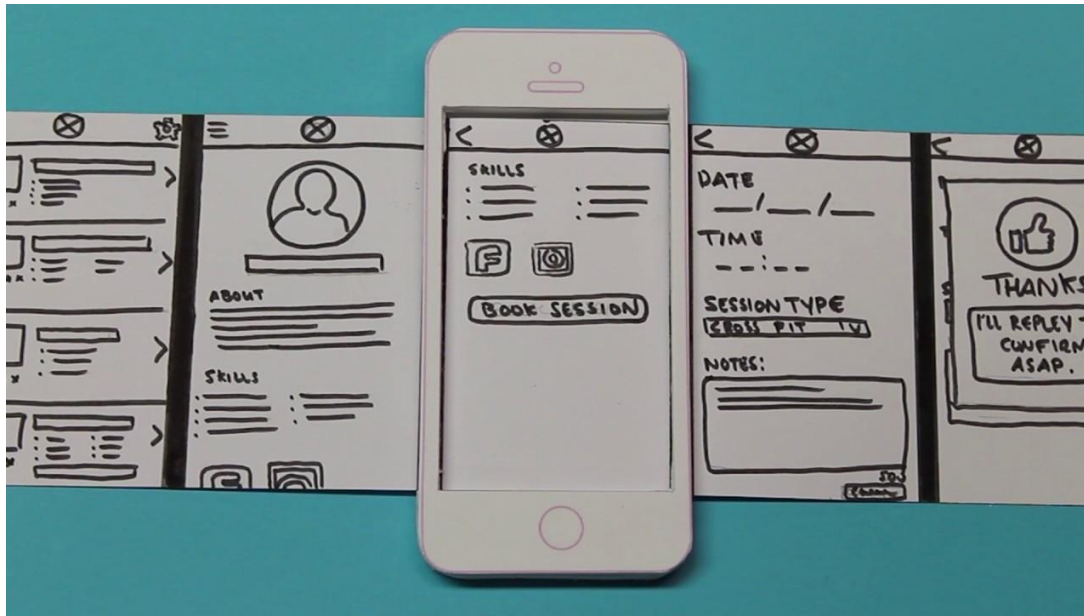


Creating Flows With Paper Prototypes



<https://youtu.be/GrV2SZuRPvo>

“Dynamic” Screens



How to Test a Paper Prototype

- The Design Team should cover these roles
- ‘Computer’ actor
 - Simulates prototype
 - Does not give any feedback that the computer would not
- Facilitator
 - Presents interface and tasks to the user
 - Encourages user to “**think aloud**” by asking questions
 - Keeps user test from getting off track
- Observer
 - Keeps mouth shut
 - Takes copious notes

<https://youtu.be/OlbdIXLunt4?t=596>

Learnable Lessons From Paper Prototypes

Can Learn

- Conceptual model
 - Do users understand it?
- Functionality
 - Does it do what's needed? Missing features?
- Navigation & task flow
 - Can users find their way around?
 - Are information preconditions met?
- Terminology
 - Do users understand labels?
- Screen contents
 - What needs to go on the screen?

Cannot Learn

- Look: color, font, whitespace, etc
- Feel: efficiency issues
- Response time
- Are small changes noticed?
 - Even the tiniest change to a paper prototype is clearly visible to user
- Exploration vs. deliberation
 - Users are more deliberate with a paper prototype; they don't explore or thrash as much

Medium Fidelity Prototypes

Wireframes, PowerPoints, Sketching tools

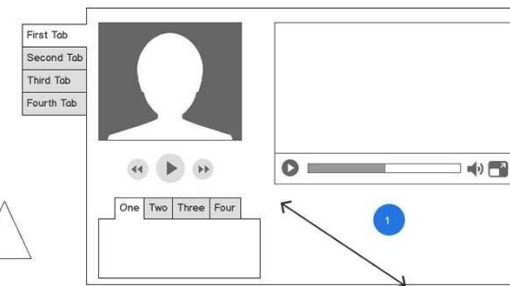
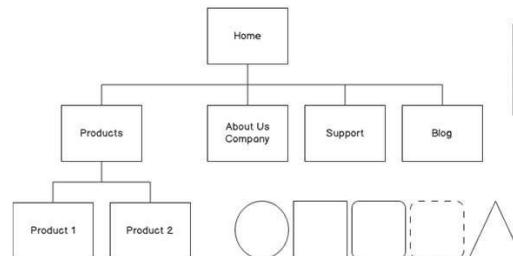
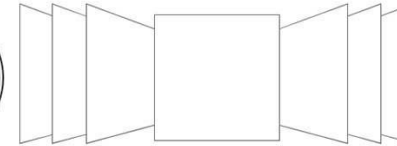
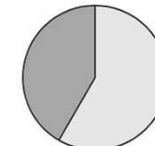
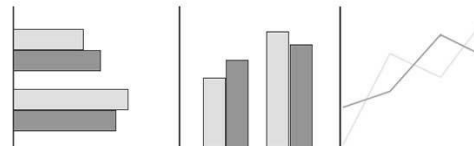
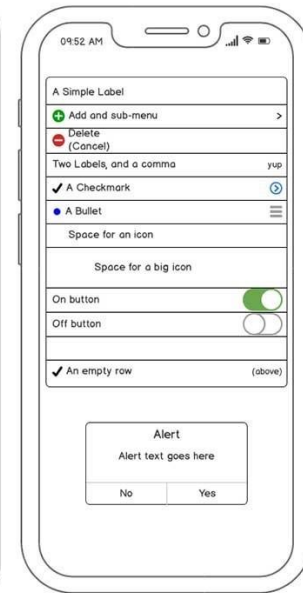
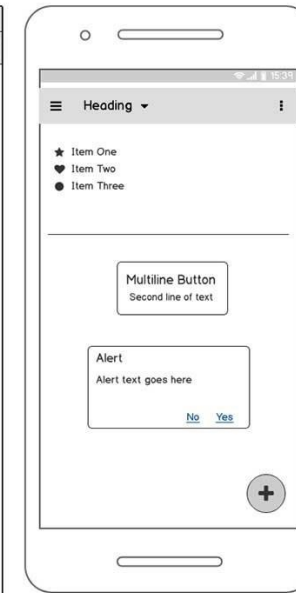
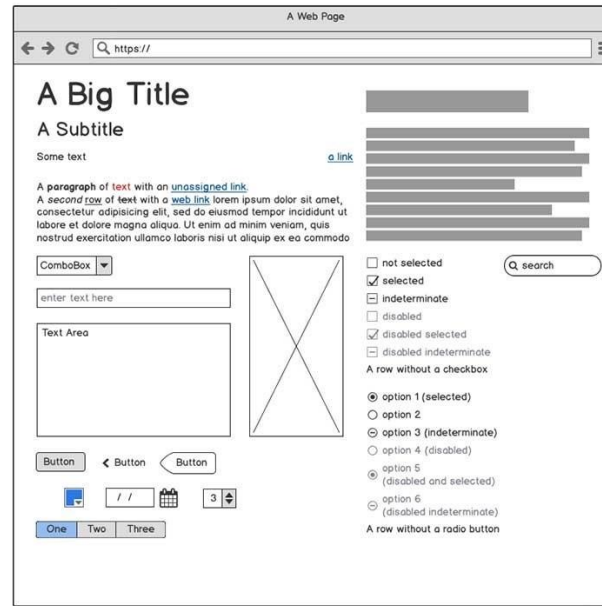
Computer Prototypes

- Interactive software simulation
 - Renders user interface
 - Accepts some user input
 - Responds by switching pages
- Medium-fidelity or High-fidelity in look & feel
- Low-fidelity in depth
 - The human operator in paper prototyping is aware of the algorithms

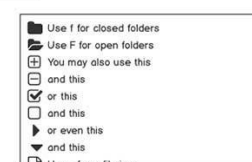
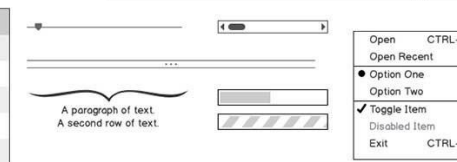
Medium-fidelity

- Also known as “Mockups” (or “Wireframe interface”)
- Design of a single screen or a set of connected screens (following a task)
- “Wavy” or “imprecise” drawing (inspired by hand drawing)
 - Want to convey the impression that the design is still preliminary
 - Mostly gray scale (or black and white)
- Usually static information (predefined pages, only)
- May suggest user device

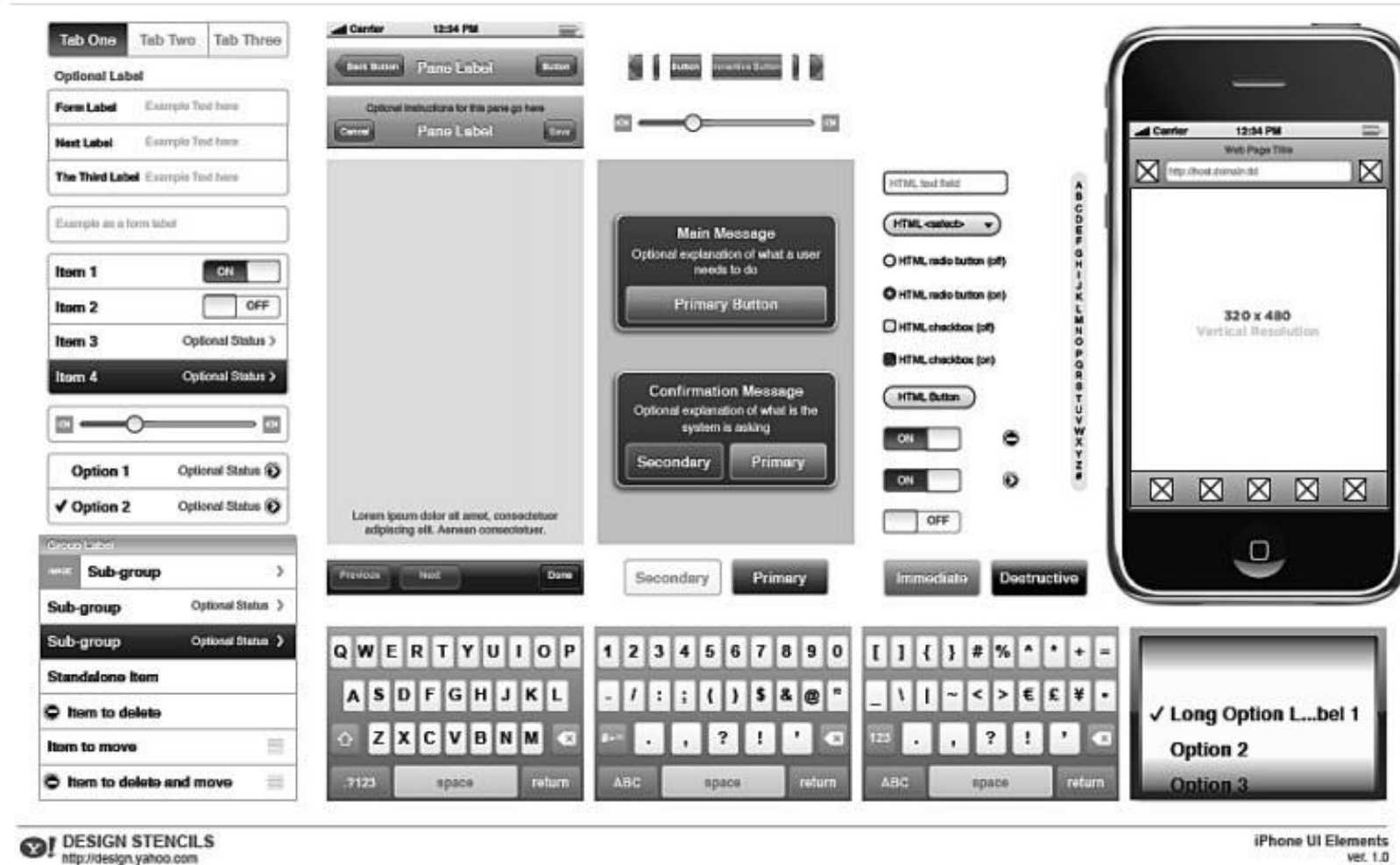
UI Design Libraries



Name (job title)	Age	Nickname	Employee
Giacomo Gullizzoni Founder & CEO	40	Peldi	☉
Marco Botton	38		☑
Tuttofare			☑
Mariah Maclochan Better Half	41	Patata	☑
Valerie Liberty Head Chef	3	Val	☑
Data Grid Docs			☐



Stencils For UI Elements



Some Tools For Med-Fidelity Prototyping



<https://www.figma.com>

<https://www.figma.com/education/>



balsamiq®

<https://balsamiq.com/wireframes/>

<https://balsamiq.cloud/>



<https://mockups.com/>

Tools' Drawbacks

- Click, not interact
 - No text entry, no data entry, no real selection of listed data
 - Widgets aren't active
- Paths are static
- The tester is engaged in a “hunt for the hotspot”, to find the (few) only widgets that really clickable
 - Good for testing understanding of the UI and the workflow
 - Not good for testing the UI behavior

High Fidelity Prototypes

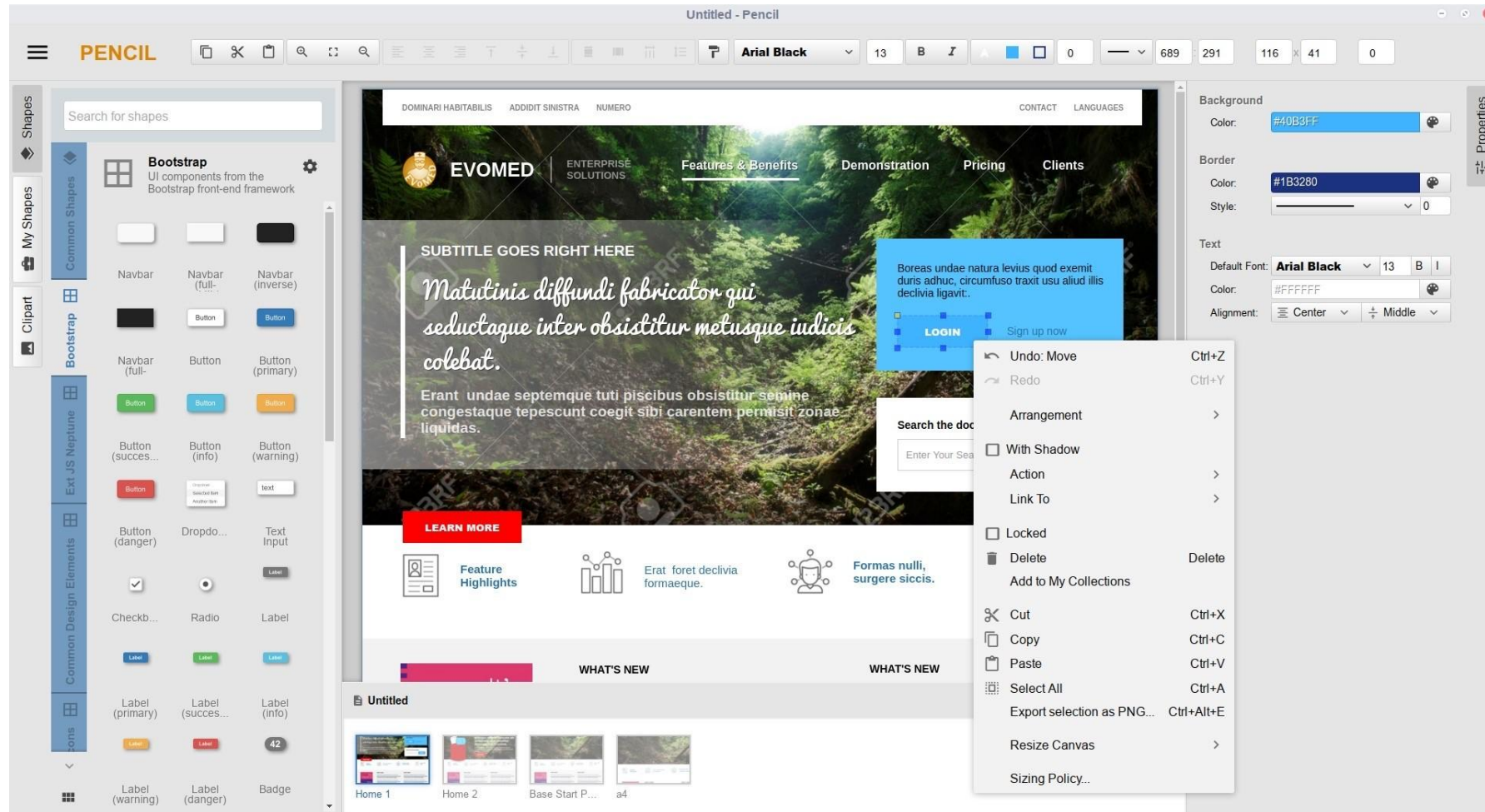
They look like the real thing. Widgets behave realistically. But it's still an illusion.

Hi-Fi Prototypes

- Actual computer application, with final-looking layout, colors, and graphics
 - May use design prototyping tools
 - May use real application code
- Much more expensive to build
- More time is spent with graphic design than interaction design
- When tested, people will mostly comment about colors, fonts, ...
 - representation communicates “finished”

High-fidelity Computer Prototypes

Semi-interactive



What Can We Learn From Hi-Fi Interactive Prototypes?

- Screen layout
 - Is it clear, overwhelming, distracting, complicated?
 - Can users find important elements?
- Colors, fonts, icons, other elements
 - Well-chosen?
- Interactive feedback
 - Do users notice & respond to status bar messages, cursor changes, other feedback
- Efficiency issues
 - Controls big enough? Too close together? Scrolling list is too long?

Suggested Video

- Prototyping: fake it till you make it
- By Apple Design Team
- <https://youtu.be/3lqh-A5Jy4Q>



Introducing

Some Tools For Interactive Hi-Fi Prototypes

No-Code



<https://www.invisionapp.com/>



<https://www.figma.com>

FROONT

<https://froont.com/>

webflow

<https://webflow.com/>



<https://principleformac.com/>

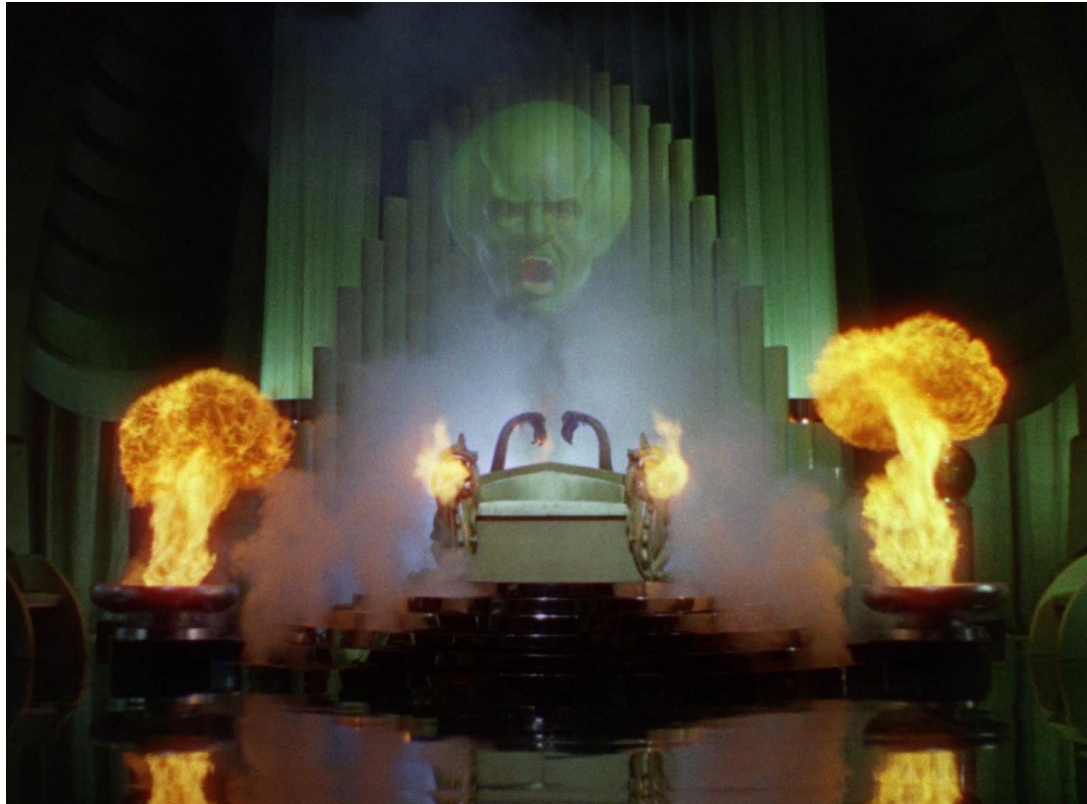
Wizard-of-Oz Techniques

Faking a technology, or filling-in for missing functionality

Goal

- How to test an application that is really complete...
 - With finalized user interface
 - With finalized algorithms
 - Also including stuff that we still are not able to implement
- ...but without actually writing the code
 - Except for a semi-interactive ‘dumb’ prototype

Remember The Man Behind The Curtain?



Remember the Mechanical Turk?



Wizard-of-Oz

- Software simulation with a human in the loop to help
- “Wizard of Oz” = “man behind the curtain”
 - Simulates the machine behavior with a human operator
 - Wizard is usually but not always hidden
- Often used to simulate future technology
 - Speech recognition
 - Learning
- Wizard may be hidden or visible
 - Must always be revealed, at least at the end

Implementing a Wizard-of-Oz Prototype

- Choose supported tasks and scenarios
- Create User Interface mock-ups
 - Implement a part of the system
 - Leave “hooks” for the Wizard’s actions
- Implement a back-office interface for the Wizard
- Define “rules of behavior” for the Wizard
 - When he should respond
 - How it should respond (the “algorithm”)

Benefits

- Faster and cheaper than most interactive prototypes
- More “real” than paper prototyping
- Creating multiple variations is easy
- Identifies bugs and issues with current design
- Can envision applications that are difficult to build
- Playing wizard allows a better understanding of algorithmic requirements

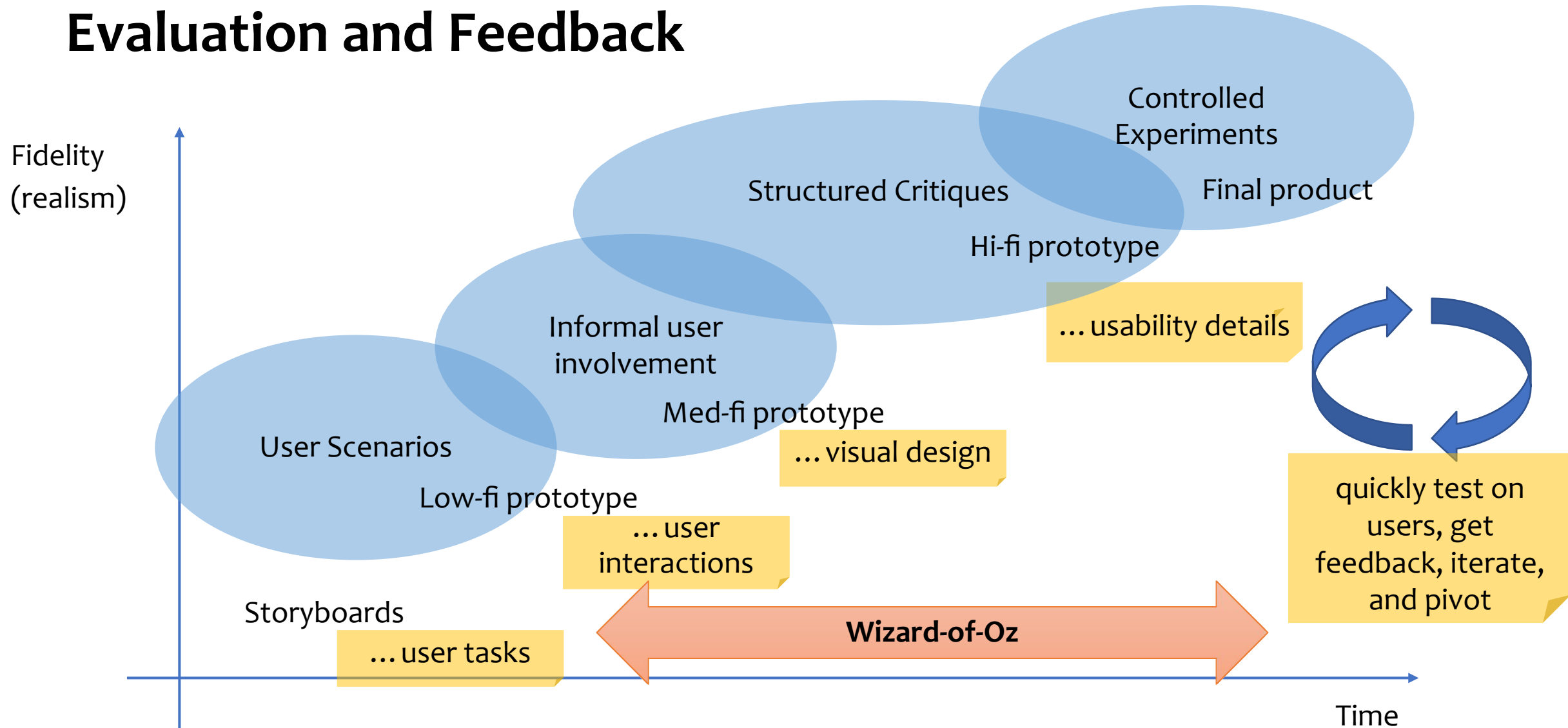
Risks

- May be over-optimistic
 - Speech recognition that always works (instead of having an error rate)
 - Super-intelligence (that will never exist)
- Wizard behavior is difficult
 - Take into account system limitations
 - Emulate expected system response
 - Within acceptable timing
- Needs at least two researchers

Wrap-up

Many different techniques, applicable to different goals and contexts

Evaluation and Feedback



References and Acknowledgments

- MIT, <https://web.mit.edu/6.813/www/sp18/classes/11-prototyping/>
- Scott Klemmer, Storyboards, Paper Prototypes, and Mockups, <https://youtu.be/z4glsttyxw8>
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